INPhO 1998

1. (a) A merry-go-round rotates with constant angular speed ω. Relative to the rotating frame of the merry-go-round, an outside object (say a man) stationary on the ground appears to revolve with the same angular speed ω in the opposite sense. Explain the motion of the object in terms of the pseudo-forces acting on it in the rotating frame of reference. **[10]**

   (b) Show that if a collision between a fast electron and an electron at rest is to lead to a production of an electron-positron pair, the kinetic energy of the fast electron must be at least $6mc^2$, where m is the mass of the electron and c is the speed of light. **[10]**

   (c) Take a system of two conductors of any shape and size with charges Q_1 and Q_2 and potentials V_1 and V_2. Let q be the charge that would flow from the conductor with the higher potential (say V_1) to the other conductor if they are shorted. Prove that the quantity $\dfrac{q}{V_1 - V_2}$ is a constant depending only on the geometry of the configuration. This quantity may be interpreted as the capacitance of the system. Does it reduce to the usual definition of capacitance when $Q_1 = Q$ and $Q_2 = -Q$? **[10]**

   (d) For a hydrogen atom, obtain the standard Bohr's formula for quantized energy levels without invoking the usual Bohr's postulates, but instead employing the following hypothesis: **[10]**

      (i) In a stationary energy state, the De Broglie wave associated with the electron forms a standing wave pattern, i.e. an integral number of wavelengths fits exactly into the circumference of the circular orbit.

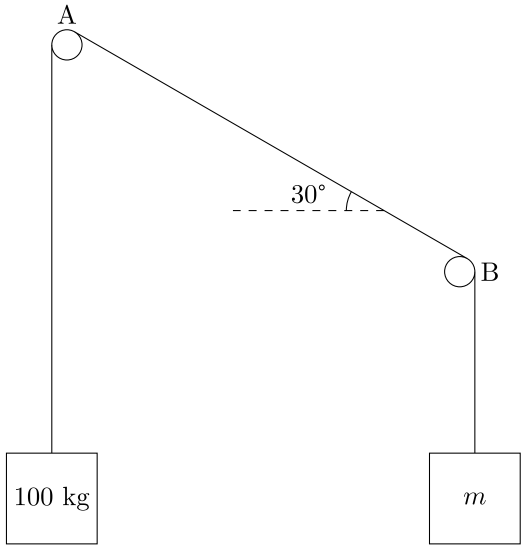
      (ii) The wavelength λ is unique for a given energy state.

   (e) The image of the objective of the telescope in the eye-piece is called the 'eye-ring'. Explain why this is the best position for viewing. Show that the angular magnification of a telescope equals the ratio of the diameter of the objective to the diameter of the eye-ring. **[10]**

2. (a) The adjacent figure shows a rope looped over two pipes A and B. If the coefficient of static friction is 0.25, determine **[15]**

      (i) the smallest and the largest values of m for which equilibrium is possible, and

      (ii) the corresponding tension in the portion AB of the rope between the two pipes.



(b) One mole of supercooled liquid water at $-15°C$ freezes irreversibly at constant one atmosphere pressure to ice at $-15°C$. Verify that the second law of thermodynamics (in entropy terms) is true for the process. **[15]**

You will need the following data:

Molar heat capacity of liquid water = 76.1 $JK^{-1}mol^{-1}$
Molar heat capacity of ice = 37.15 $JK^{-1}mol^{-1}$
(Neglect variations in molar heat capacity with temperature)
Latent heat of fusion of water = 333.5 Jg^{-1}
Molecular mass of water = 18.02 g mol^{-1}

Assume that the metastable supercooled liquid water state is an equilibrium state and that the surroundings remain at equilibrium at $-15°C$.

3. **Propagation of Electromagnetic Waves in the Ionosphere**

We wish to understand why waves of right-handed and left-handed circular polarizations propagate differently in the ionosphere. To model the situation in a simple way, we start with some rough data, make approximations, and proceed step by step as follows:

(a) The ionosphere is a plasma with free-electron density $n = 10^{11}$ electrons/m^3 and a static uniform magnetic field $\vec{B}_0$ of magnitude 3×10^{-5} T. **[8]**

Consider a transverse electromagnetic wave of frequency ω propagating in the direction of $\vec{B}_0 = B_0\hat{k}$. Take the wave to be left circularly polarized. (This means that the electric field vector $\vec{E}$ of the wave has a constant magnitude E_0 but rotates in a circle with frequency ω in an anticlockwise sense, when looking into the wave. The right circularly polarized wave is similar except that the sense of rotation of the electric field vector is clockwise, when looking into the wave). Make the following approximations:

(i) The static field B_0 is much stronger than the B of the transverse wave.
(ii) The amplitude of electronic motion is much smaller than the wavelength of the wave.
(iii) Collisions of electrons can be neglected.

Show that the equation of electronic motion under these approximations can be written as:

$$m\frac{d^2r}{dt^2} = 2\frac{dr}{dt} \times B_0 - eE_0(\hat{\imath}\cos\omega t + \hat{\jmath}\sin\omega t)$$

where $\hat{\imath}, \hat{\jmath}, \hat{k}$ are mutually perpendicular unit vectors of a right-handed co-ordinate system of axes (m = mass of electron; $-e$ = charge of electron).

(b) Show that this leads to a dipole moment for each electron given by

$$p = -\frac{e^2}{m\omega(\omega - \omega_0)}E \quad \text{where} \qquad \omega_0 = \frac{eB_0}{m}$$

Interpret ω_0.

(c) Repeat the analysis in (a) and (b) above for the right circularly polarized wave, and show that the dielectric constant of the medium for the two types of waves is **[8]**

$$K_{\mp} = 1 - \frac{\omega_p^2}{\omega(\omega \mp \omega_0)}, \text{ with } \qquad \omega_p^2 = \frac{ne^2}{m\varepsilon_0}$$

where the upper sign corresponds to the left-circularly polarized wave and the lower sign for right circularly polarized wave.

(d) Obtain numerically the values of ω_p and ω_0 **[8]**

$$\varepsilon_0 = 8.85 \times 10^{-12}\,\mathrm{C^2\,N^{-1}\,m^{-2}},$$
$$e = 1.6 \times 10^{-19}\,\mathrm{C}, \qquad m = 9.1 \times 10^{-31}\,\mathrm{kg}.$$

Show that for $\omega = 10^6$ Hz, one of the two types of waves above will be totally reflected by the ionosphere. Specify which?

(e) How do your conclusions change if the direction of propagation of the wave is antiparallel to $\vec{B}$? **[4]**